

Exploring COVID-19 Trends: Unwrapping Insights related to COVID-19 Pandemic across the United States



UNIVERSITY
OF WOLLONGONG
AUSTRALIA



Prepared by:
Cindy Chiandary - 7406927

Assessment 2

MBAS901 - Essential Elements
for Business Analytics

Lecturer: Dr Dinindu Koliya
Wedanage

Trimester 3 2024



Table of Contents

Table of Contents	2
Introduction.....	3
Dataset Description	4
Overview	4
Categorical Variables.....	4
Measures	4
Descriptive Statistics	5
Overall Aim.....	6
Hypotheses	7
Methodology	8
Data Pre-Processing	8
Charts.....	8
Results and Discussion.....	9
Conclusion	25
Summary.....	25
Key Insights	25
Recommendations	25
Reference List.....	27
Appendix.....	29

Introduction

The rapid spread of the COVID-19 virus has had an immense impact on daily activities, economy of countries, groups, and individuals, as well as global health. The pandemic originated in Wuhan, China in late 2019 and began to spread around the globe in early 2020, including the United States (BBC 2020; BBC 2022; Hale 2023). As a result, the US government and public healthcare have been closely monitoring the spread of the virus to prevent the further transmission and minimise its negative impacts (World Health Organization 2020; World Health Organization 2021). Moreover, the New York Times (NYT) had provided comprehensive daily counts of COVID-29 cases and deaths across the US between January 2020 and March 2021, alongside played a crucial role in collecting the data linked to the pandemic (SAS Viya for Learning 2024). This dataset showcased a thorough understanding of the pandemic's evolution throughout time and across different areas of the United States.

As a data analyst for a public health institution, I will analyse the COVID-19 dataset from NYT to obtain insightful information through exploratory analysis. The goal of this exploratory analysis is to gain key insights that are essentials in presenting useful information on the COVID-19 pandemic to the public healthcare. To obtain the better insights, I will pre-process the data as necessary and utilise the best-fit charts to effectively communicate the information based on the dataset. By the time this analysis is done, I will receive a better understanding of the dynamics of the pandemic and support the country's initiatives to mitigate the outbreak and its effects on the health of US citizens, economic sectors, and public activities.

Dataset Description

Overview

The Corona Virus dataset provided by NYT generally gather the data for the cases and fatalities related to the COVID-19 pandemic across the United States. The data collection on this dataset is based on daily calculation of COVID-19 confirmed cases and deaths. There are four classifications included on this dataset: category and measure, alongside the inclusion of 24 columns (*refer to Data Dictionary on Appendix*).

Categorical Variables

Category	Data Type	Descriptions
County	String	List of counties across the US.
Date by Month and Year	Date	Dates for the update of COVID-19 cases and deaths, recorded in MM-YYYY format.
State	String	The list of 50 US states, capital of the US, and five other US-controlled territories.
Country_id	String	The unique identifier assigned to countries.

Measures

Measures	Aggregation	Data Type	Descriptions
Cases	Sum Average Minimum Maximum Standard Deviation Standard Error	Integer	Confirmed cases of people who have tested positive for COVID-19 on the given date.
County Latitude	Sum	Float	The county's latitude coordinate.
County Longitude	Sum	Float	The county's longitude coordinate.
Deaths	Sum Average Minimum Maximum	Integer	The confirmed deaths caused by COVID-19 on the given date.

	Standard Deviation Standard Error		
Frequency	Sum		
State Latitude	Sum	Float	The state's latitude coordinate.
State Longitude	Sum	Float	The state's longitude coordinate.

Descriptive Statistics

Key Variables	Distinct Values	Mean	Median	Minimum	Maximum	Standard Deviation
Date	427	2020/09/25	2020/09/27	2020/01/25	2021/03/22	103.9511
Cases	44,594	3358.475	440	0	1,215,330	17,079.32
Deaths	4,485	74.68631	8	0	30,564	521.1533

(Sas Viya for Learning 2024)

Overall Aim

The main objective of this analysis is to build information regarding COVID-19 based on received key insights, such as trends and patterns related to COVID-19 cases and deaths across the United States. Specifically, this analysis aims to:

1. Identify the trends and patterns from the COVID-19 dataset through useful charts and graphs.
2. Gain the key insights.
3. Interpret the insights through storytelling.
4. Compare the total and average daily cases and death toll by states.
5. Suggest effective methods to preventing the virus spreading further to decrease the number of COVID-19 cases and fatalities.

The main purposes of the specific aims above are to provide insightful information regarding the COVID-19 outbreak in the United States, and to inform the public health care about the dynamic of the pandemic. Hence, the public health care will be able to make informed decisions in minimising the negative impacts of the pandemic, alongside preventing the further transmission of the virus based on the insightful information made from the dataset.

Hypotheses

The five hypotheses below are to be answered on this report:

Hypotheses 1: The higher rates of the COVID-19 cases, the higher rates of the causalities caused by COVID-19 across the United States.

This hypothesis aims to determine whether the COVID-19 related mortality in the US will follow in proportion the number of confirmed COVID-19 cases.

Hypotheses 2: The number of COVID-19 cases and deaths on each state represents the average rate of COVID-19 cases and deaths.

The second hypothesis is to test whether the average rate of confirmed cases and fatalities per state comes after the total number of COVID-19 and deaths by state.

Hypotheses 3: States with more counties have higher COVID-19 total and average cases.

The third hypothesis seeks to test whether the states with higher number of counties are more likely to report higher number of COVID-19 cases and deaths.

Hypotheses 4: The cases and deaths due to the COVID-19 in US-territories are more uncommon compared to the 50 US states.

The purpose of this hypothesis is to further prove whether COVID-19 cases and deaths are rare in five US-territories.

Hypotheses 5: There will be a significant increase of COVID-19 cases over the next 12 months.

The final hypothesis aims to predict the trends and patterns of COVID-19 within the next 12 months.

Methodology

Data Pre-Processing

This report used the following data pre-processing methods:

1. Transform the date format into MM-YYYY.
2. Remove (hide) the repeated data (e.g. state, and county).
3. Add the data with another sum (e.g. add the cases and deaths in the “average” sum).

Charts

This analysis utilise the charts below:

1. Correlation Matrix

To determine the correlation between COVID-19 cases and deaths around the US, alongside finding the relationship between COVID-19 cases and counties latitude and longitude.

2. Comparative Time Series

To identify the trends and patterns of COVID-19 cases and deaths between January 2020 and March 2021.

3. Dual Axis Bar Line Chart

To further prove the correlation between the total COVID-19 cases and deaths.

4. Dual Axis Line Chart

To substantiate the link between the average COVID-19 cases and death tolls.

5. Dual Axis Bar Chart

To develop **Hypothesis 2** that aims to ascertain the link between the total and average COVID-19 cases, alongside the total and average causalities related to COVID-19.

6. Bar Charts

To rank the states based on the highest and lowest number of COVID-19 cases and deaths.

7. List Table

To categorise the counties with highest number of COVID-19 cases and deaths based on the US state.

8. Time Series Plot

To identify the COVID-19 trends across the US.

9. Forecast

This is applicable for **Hypothesis 5**, which aims to predict the number of COVID-19 cases in the US within the next 12 months.

Results and Discussion

In general, I assume that there is a relationship between the cases and fatalities of COVID-19. By data analysis from SAS Viya, I found that there is a relation between COVID-19 cases and deaths, as evidenced by the correlation of total and average COVID-19 cases and fatalities, scored at 0.7833. The following figures below indicates an increase of likelihood of increased causalities with an increase in COVID-19 patients in the US (Sas Viya for Learning 2024).

Correlation of COVID-19 Cases and Deaths



Figure 1.1

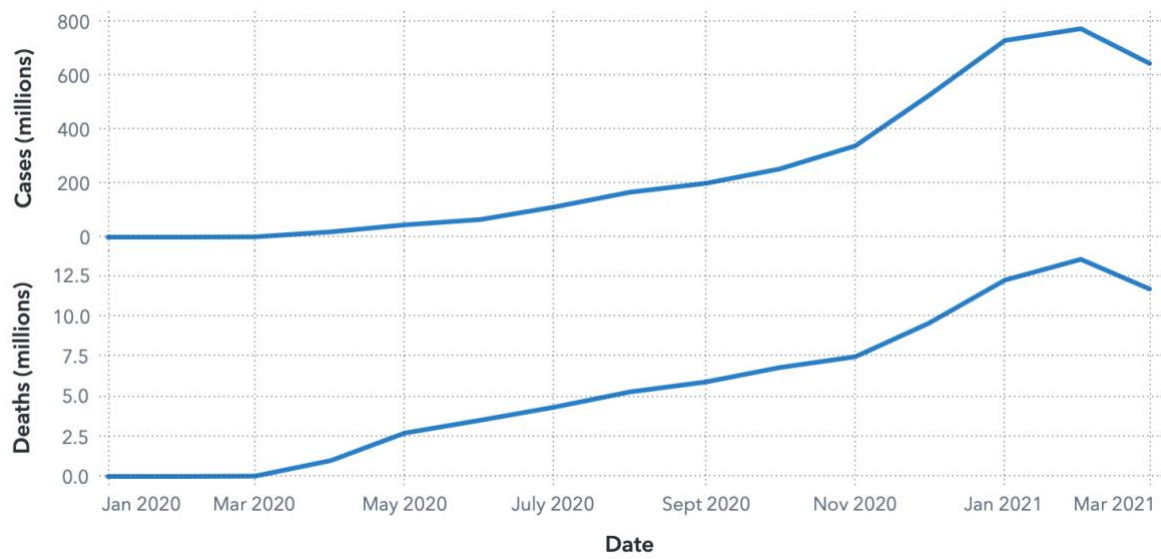


Figure 1.2

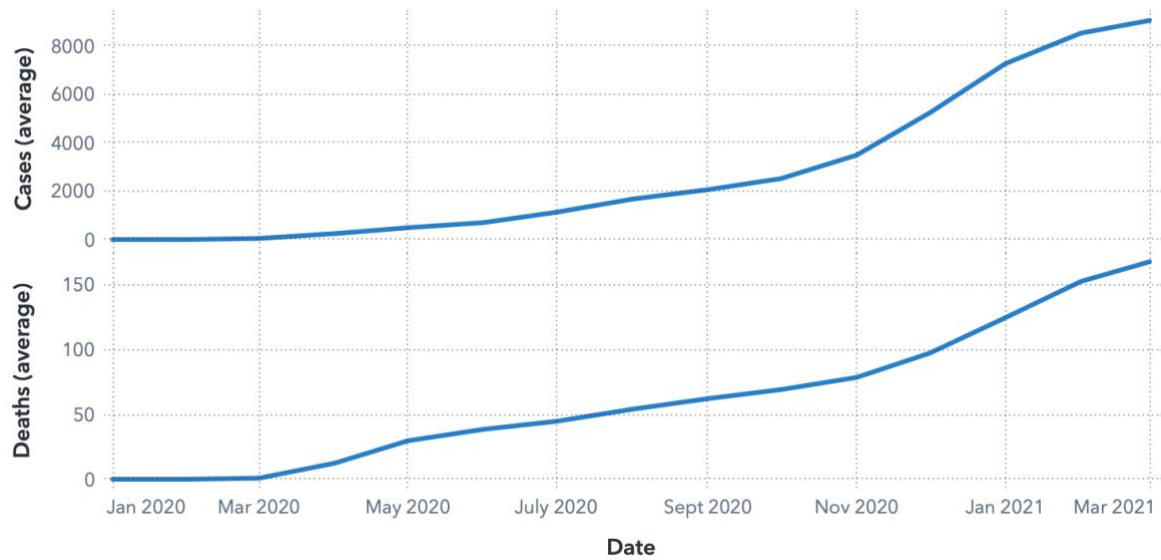


Figure 1.3

Figures 2.1 and 2.2 below show the similarity between the COVID-19 cases and deaths rates across many states and US territories, which further supports this viewpoint (Sas Viya for Learning 2024).

Total COVID-19 Cases and Deaths by State

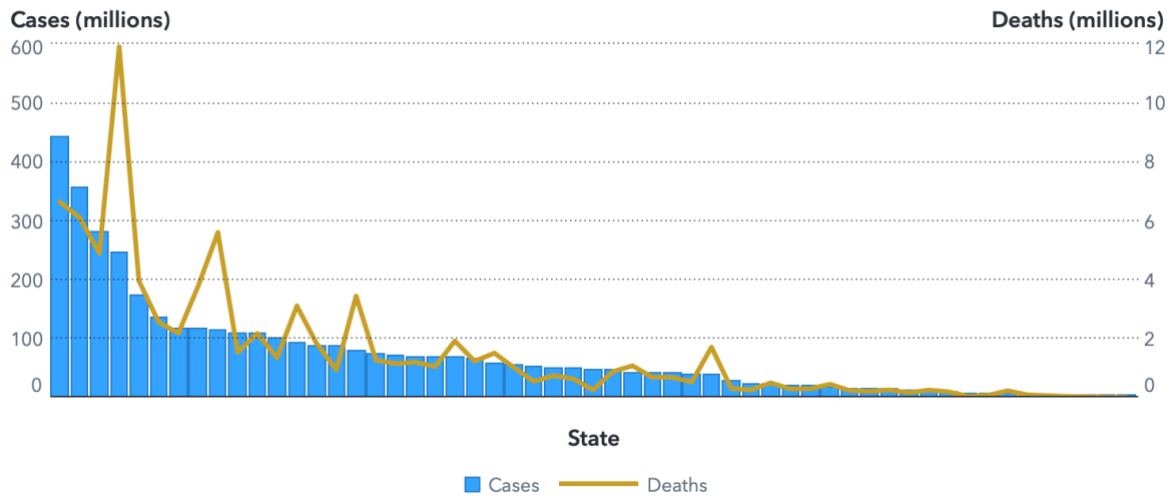


Figure 2.1

Total COVID-19 Cases and Deaths by State

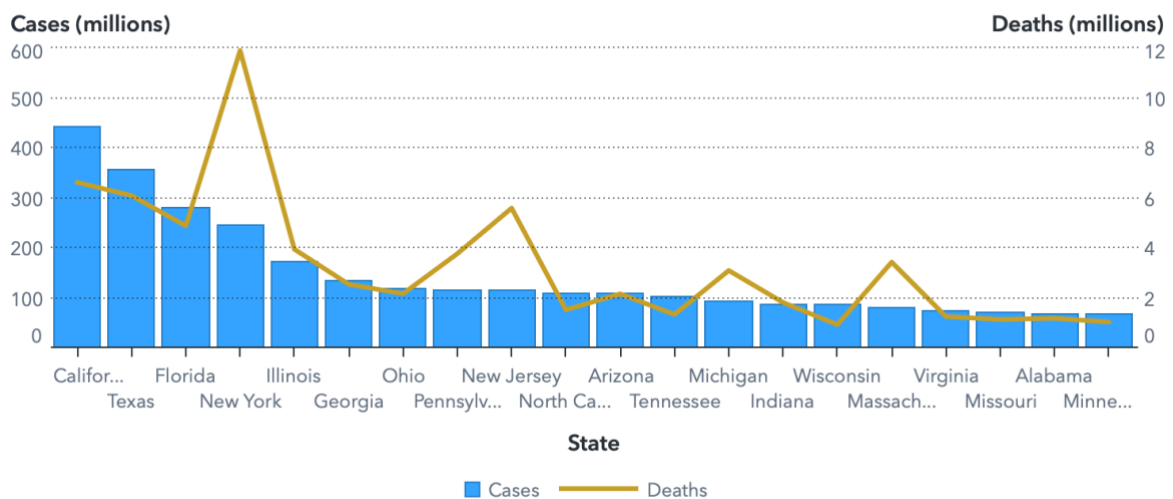


Figure 2.2

Similarly, the greater number of COVID-19 cases is associated with the higher rate of mortality. Figures 2.3 and 2.4 below illustrate this by showing how closely spaced apart the average cases and mortality are from one another. This strengthens the correlation between COVID-19 average cases and fatalities in the most states and US-controlled territories (Sas Viya for Learning 2024).

Average COVID-19 Cases and Deaths by State

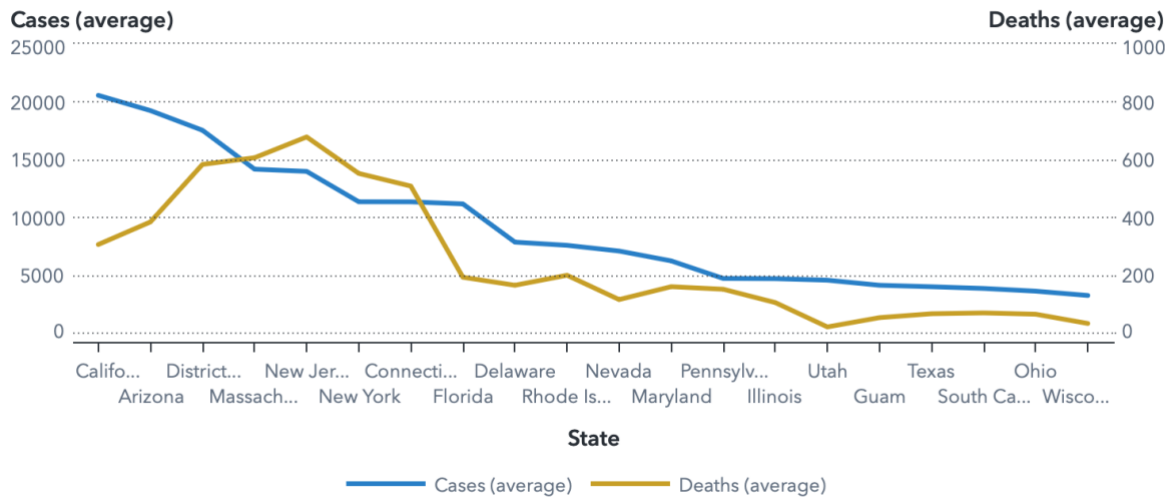


Figure 2.3

Average COVID-19 Cases and Deaths by State

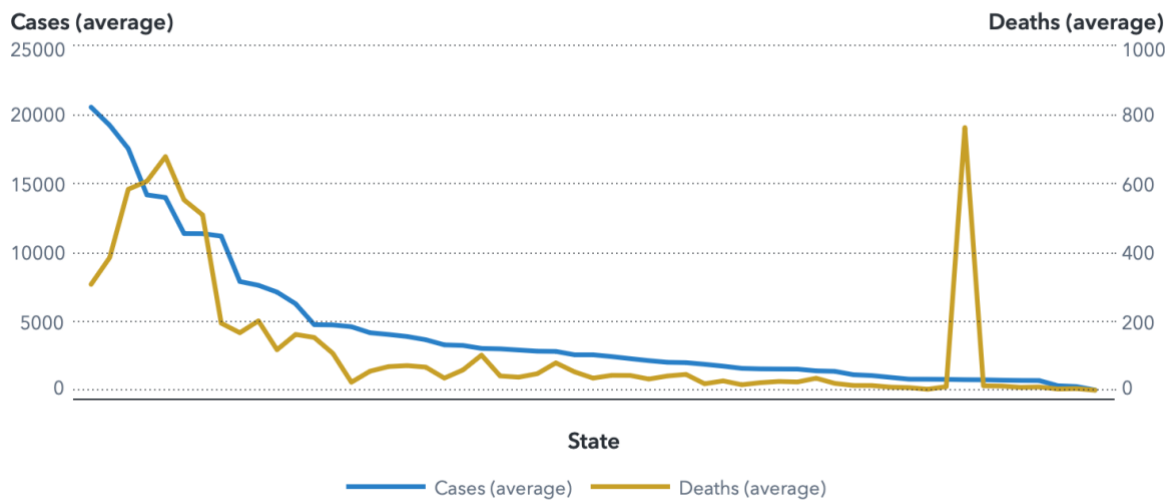


Figure 2.4

Overall, the correlation matrix, alongside the graphs above demonstrate the high association between the cases and deaths related to COVID-19. This suggests that the higher the confirmed cases, the higher the casualties and vice versa.

Secondly, there might be corresponding between the total and average COVID-19 cases and casualties. However, the graphs below showcases that the average COVID-19 cases seldom correspond with the total COVID-19 cases, even though in certain states, they may be corresponding to each other (Sas Viya for Learning 2024).

Total and Average COVID-19 Cases by State

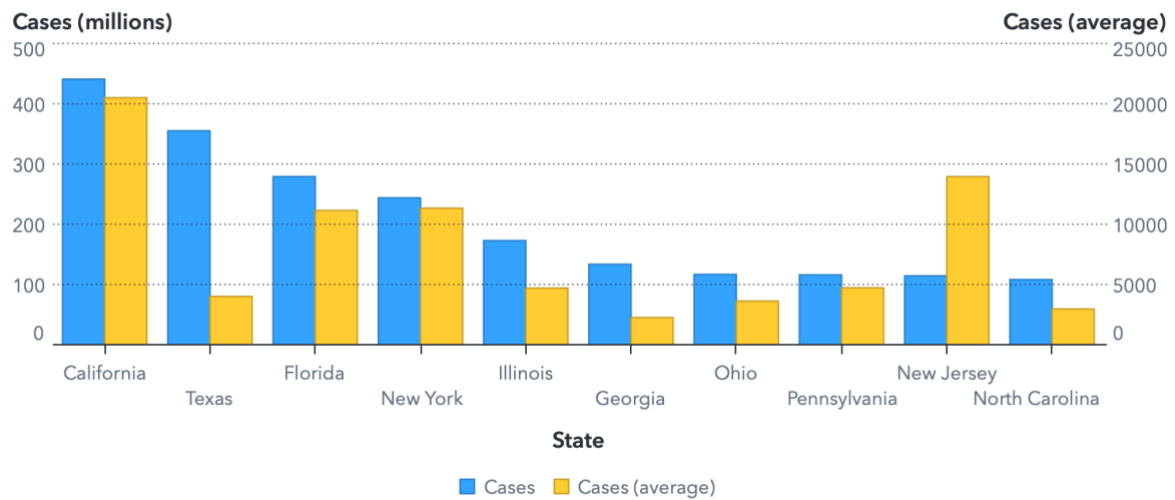


Figure 3.1

Total and Average COVID-19 Cases by State

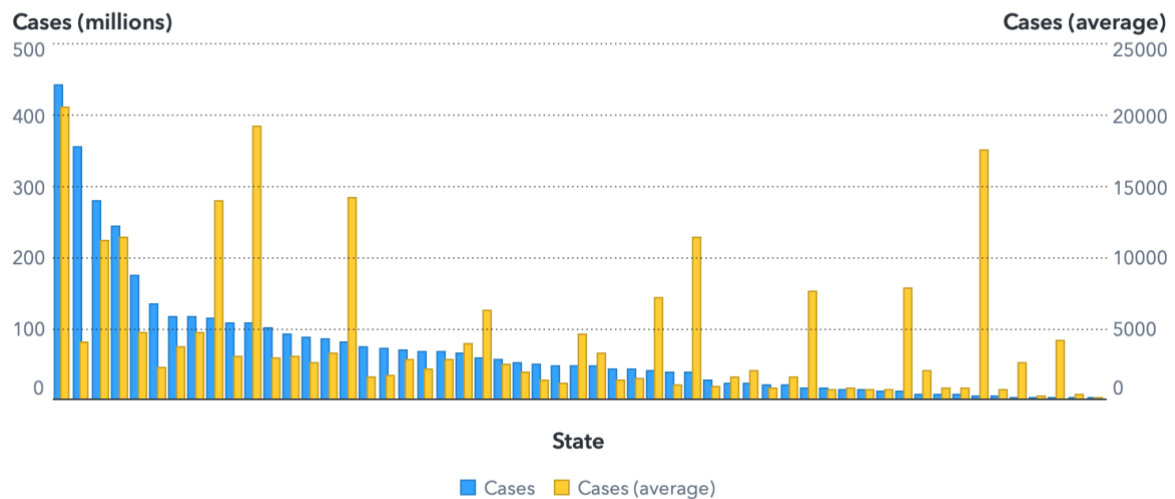


Figure 3.2

Moreover, the average deaths do not follow the rate of total deaths, although some states may present the similar rate between the total and average mortality rate (Sas Viya for Learning 2024).

Total and Average Deaths related to COVID-19 by State

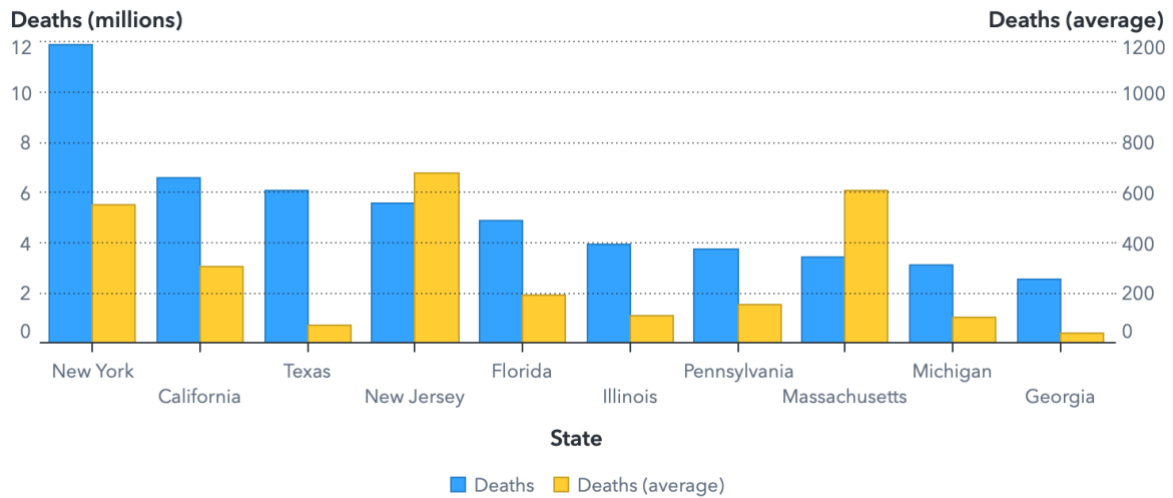


Figure 3.3

Total and Average Deaths related to COVID-19 by State

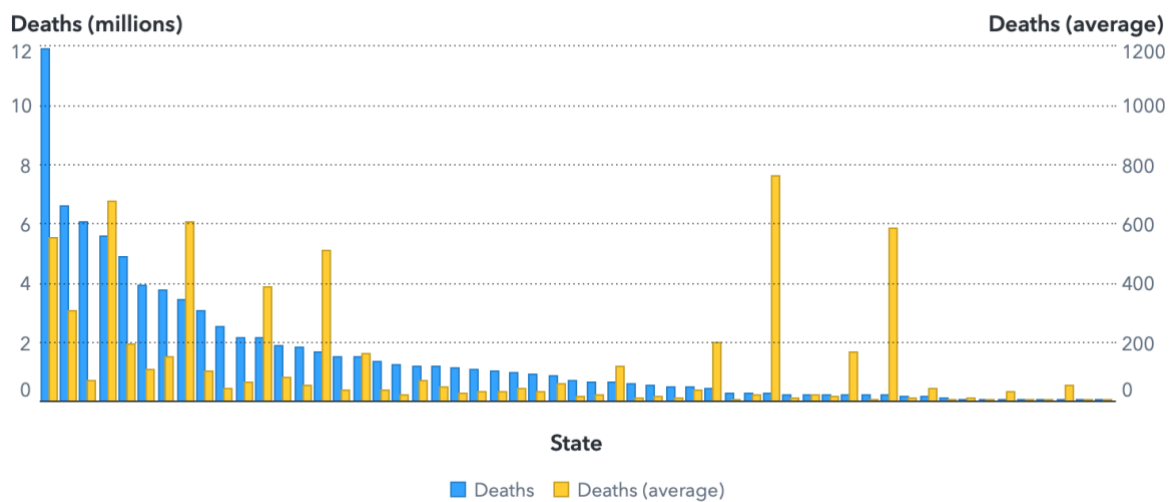


Figure 3.4

Thereby, the overall cases and death tolls of COVID-19 do not reflect the average rate on each state, which leads this analysis to include both the total and average sum of the dataset related to COVID-19 cases and deaths.

Thirdly, it is assumed that the reported cases per state may be higher due to the share counts from counties on each state. Thus, states with higher total and average cases may have more counties. However, this is not true since only four out of ten states with highest COVID-19 total cases have the highest numbers of counties. Also, the fact that there are no states with top ten highest average COVID-19 cases are listed on the states with most

counties weakened this assumption (Wisevoter 2023; Sas Viya for Learning 2024; World Population Review 2024) (*refer to Figures 4.1 and 4.2 below*).

Cases by State

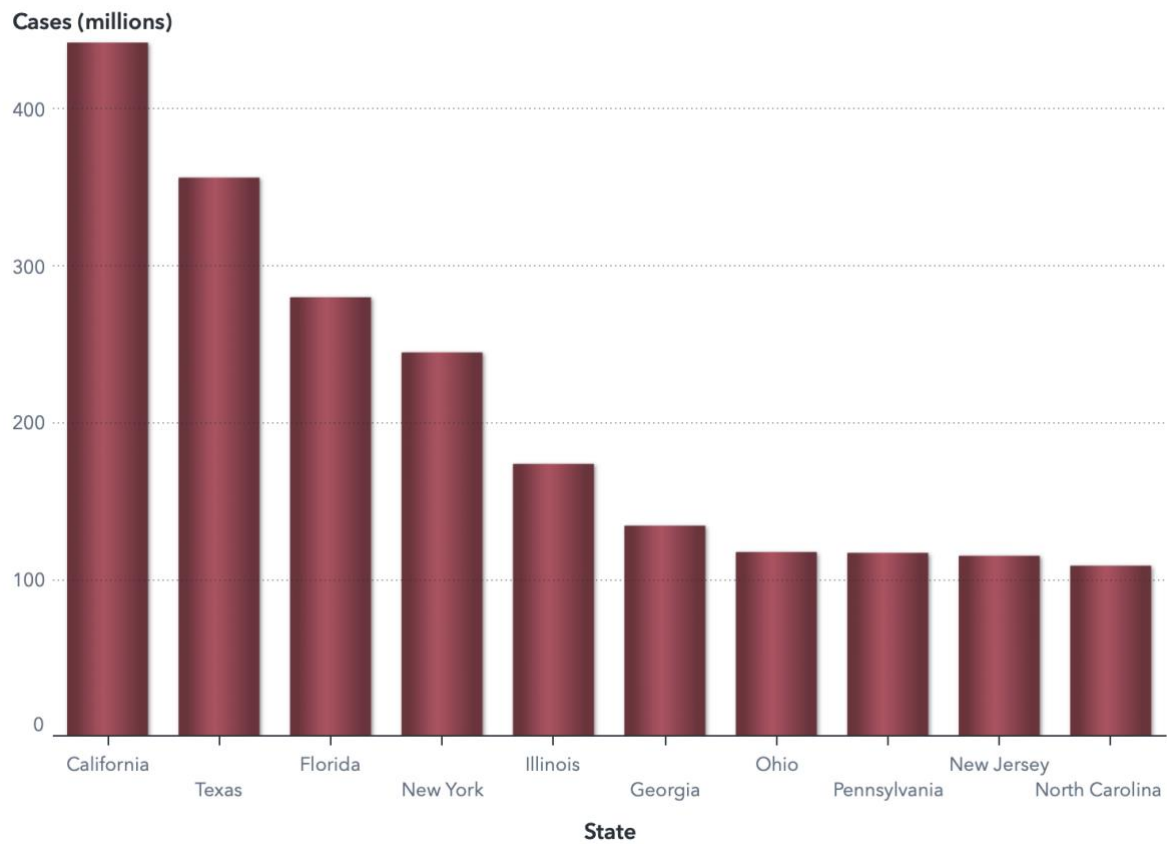


Figure 4.1

Cases (average) by State

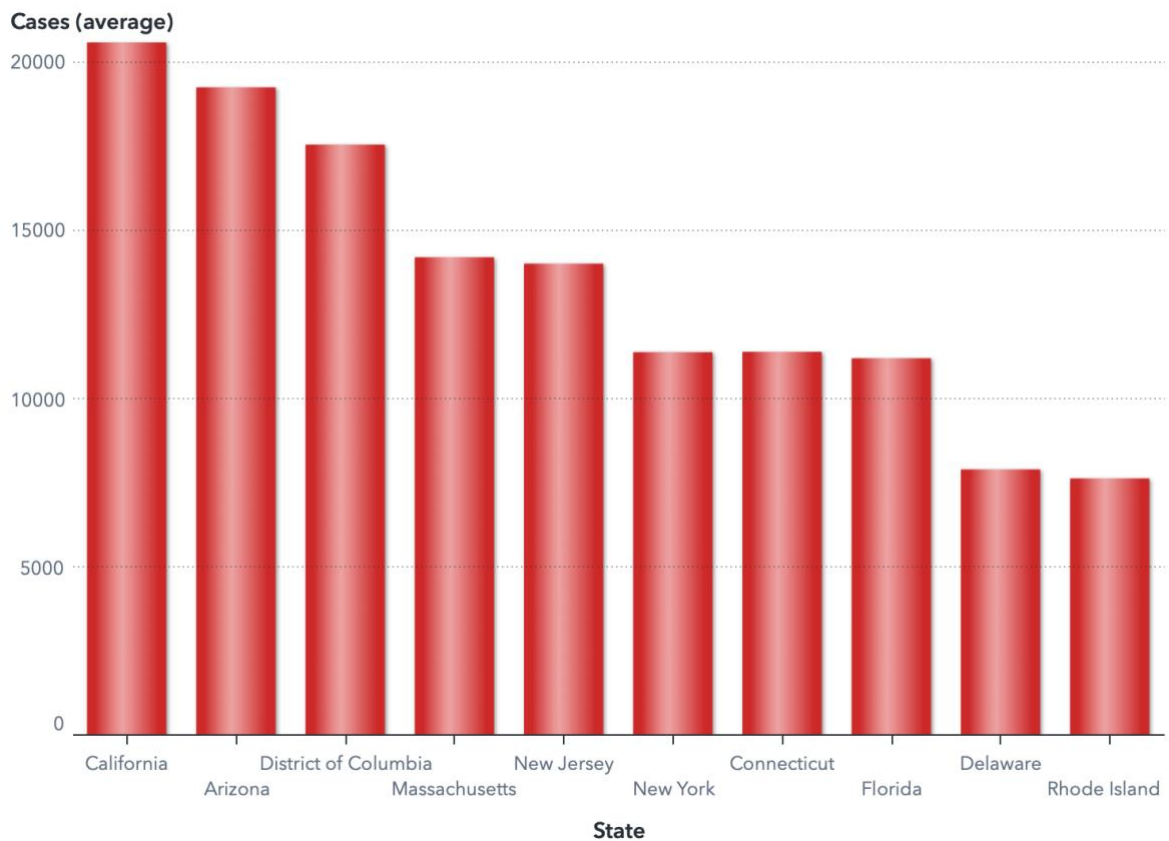


Figure 4.2

Moreover, counties that reported highest total and average COVID-19 cases are mostly located in the states that are not ranked as “US states with most counties” (see Figure 4.3 and 4.4). The tables below contradict the link between the number of counties on each state and the total and average of COVID-19 cases in the US (Sas Viya for Learning 2024).

County	State	Cases ▼
Los Angeles	California	149496491
New York City	New York	117844211
Cook	Illinois	76294217
Maricopa	Arizona	68145763
Miami-Dade	Florida	63900815
Harris	Texas	51818112
Dallas	Texas	38943760
Clark	Nevada	32009112
Orange	California	31320725
Orange	Florida	16469464

Figure 4.3

County	State	Cases (average) ▼
Los Angeles	California	354257.08768
New York City	New York	304507.00517
Cook	Illinois	179939.19104
Miami-Dade	Florida	169498.18302
Maricopa	Arizona	161482.85071
Harris	Texas	135295.33159
Dallas	Texas	103025.82011
Clark	Nevada	83574.704961
Orange	California	74044.267139
Orange	Florida	43918.570667

Figure 4.4

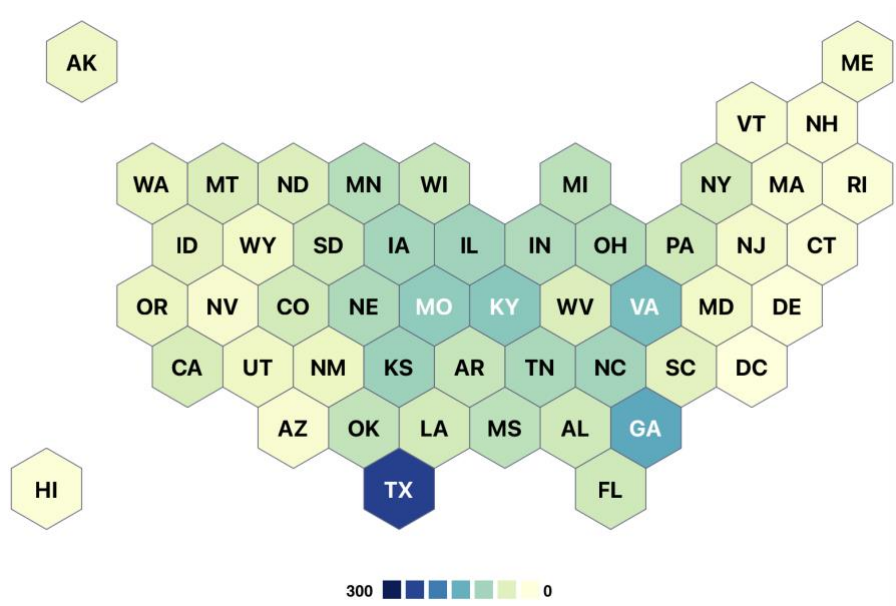
In addition, Figures 4.5 and 4.6 also showcases the contradiction of the statement “the higher the number of counties, the higher the COVID-19 related death tolls on each state”. This shows that the number of counties does not reflect the number of COVID-19 cases and deaths. Hence, the number of counties does not determine the cases of COVID-19 (Sas Viya for Learning 2024).

County	State	Deaths ▼
New York City	New York	8234169
Los Angeles	California	2777628
Cook	Illinois	2037776
Maricopa	Arizona	1247243
Miami-Dade	Florida	990354
Harris	Texas	801717
Clark	Nevada	541385
Orange	California	494075
Dallas	Texas	464758
Jefferson	Louisiana	203462

Figure 4.5

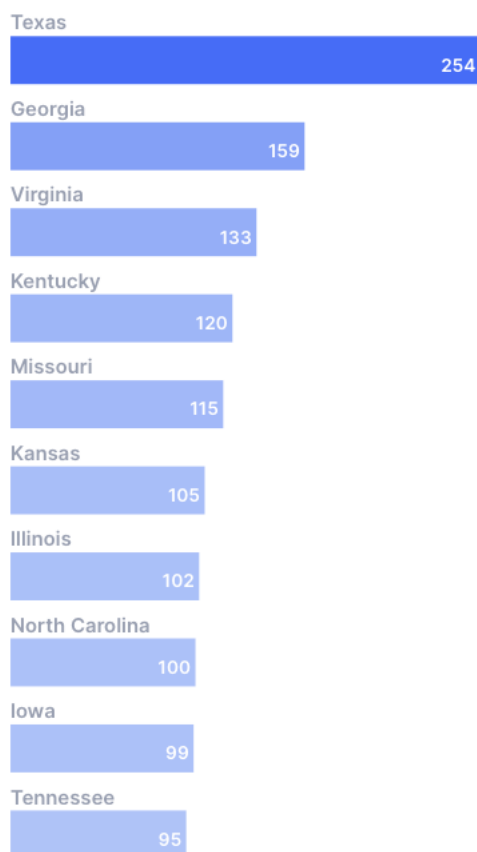
County	State	Deaths ▼
New York City	New York	8234169
Los Angeles	California	2777628
Cook	Illinois	2037776
Maricopa	Arizona	1247243
Miami-Dade	Florida	990354
Harris	Texas	801717
Clark	Nevada	541385
Orange	California	494075
Dallas	Texas	464758
Jefferson	Louisiana	203462

Figure 4.6



States with Most Counties (World Population Review 2024)

Total Number of Counties ?



(Wisevoter 2023)

According to the COVID-19 dataset provided by NYT, the COVID-19 outbreak more likely affected the US states compared to any of the five US-territories. However, this does

not imply that COVID-19 are rare in US-territories. As shown in Figures 5.1 and 5.2, there are only three territories with ten rarest COVID-19 cases by state (Sas Viya for Learning 2024).

Top 10 States with Least COVID-19 Total Cases

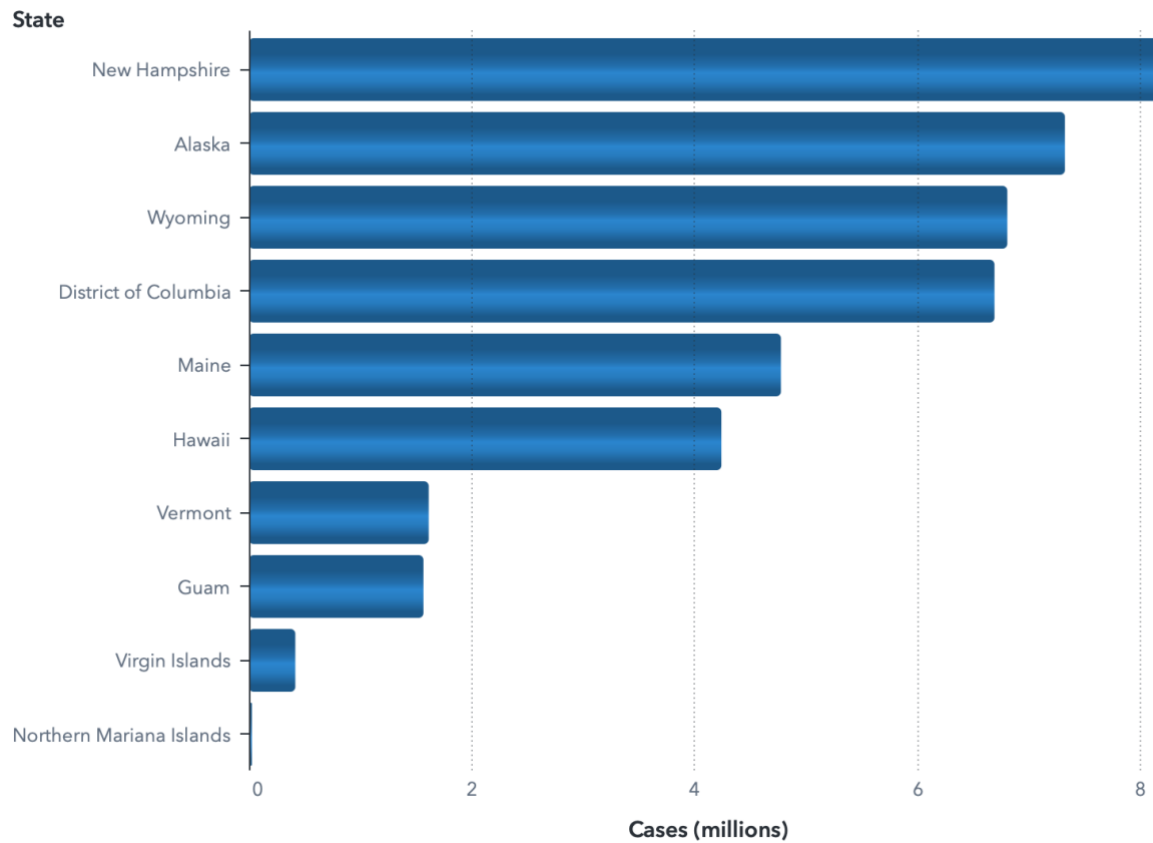


Figure 5.1

Top 10 States with Least COVID-19 Average Cases

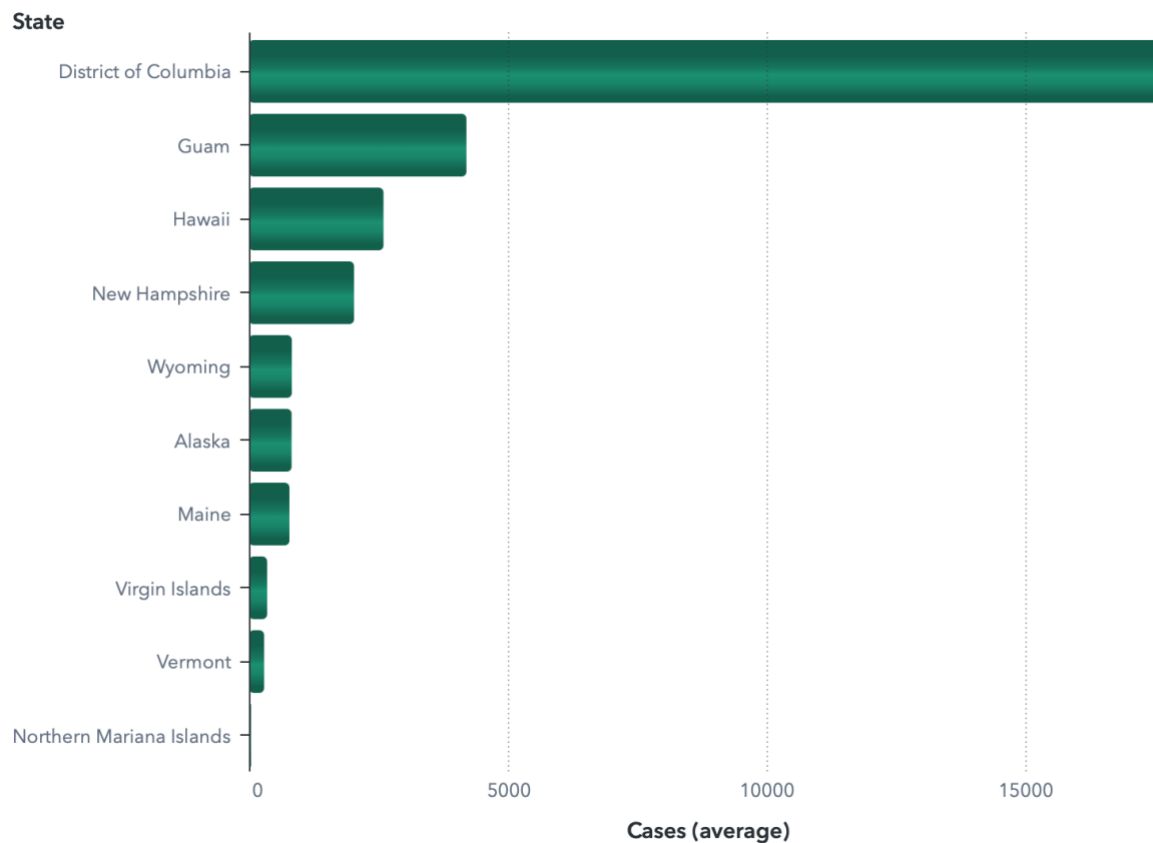


Figure 5.2

Comparably, Figure 5.3 reveals that only three territories had the lowest COVID-19 deaths, alongside Figure 5.4 that demonstrates that two out of five US-territories have the lowest average deaths related to COVID-19. Thus, simply because COVID-19 cases were more numerous in US states does not prove that COVID-19 is uncommon in all US-territories (Sas Viya for Learning 2024).

Top 10 States with Least Total Deaths related to COVID-19

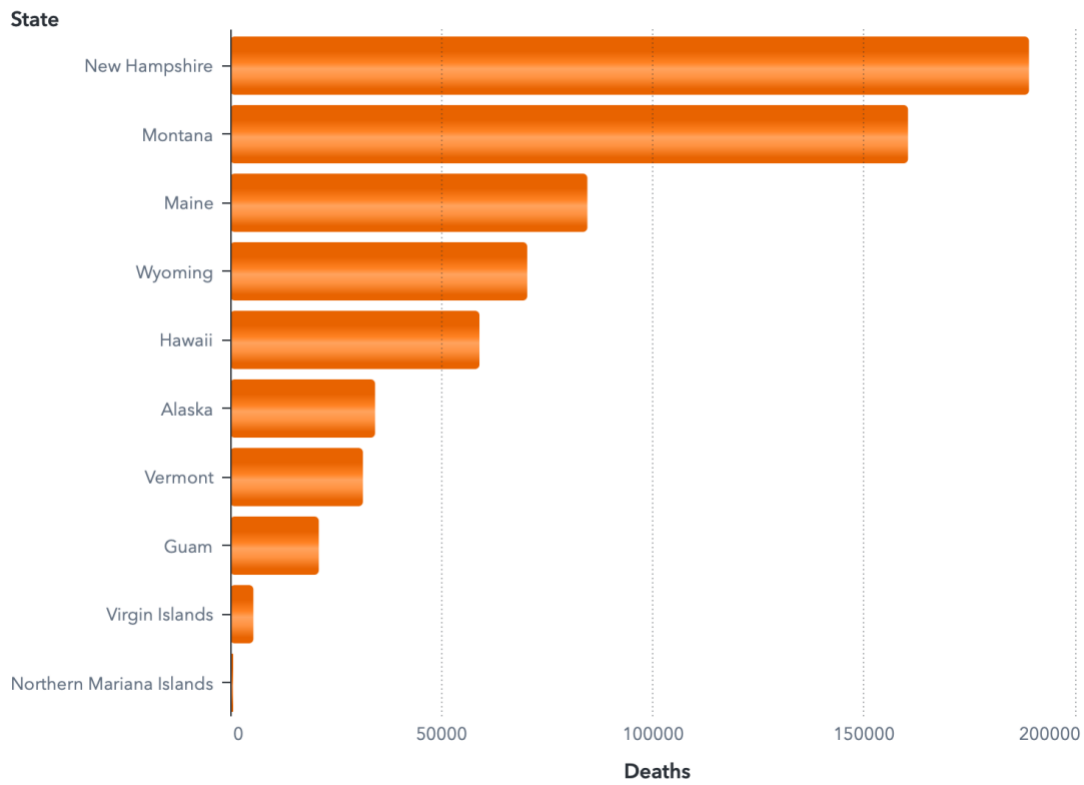


Figure 5.3

Top 10 States with Least Average Deaths related to COVID-19

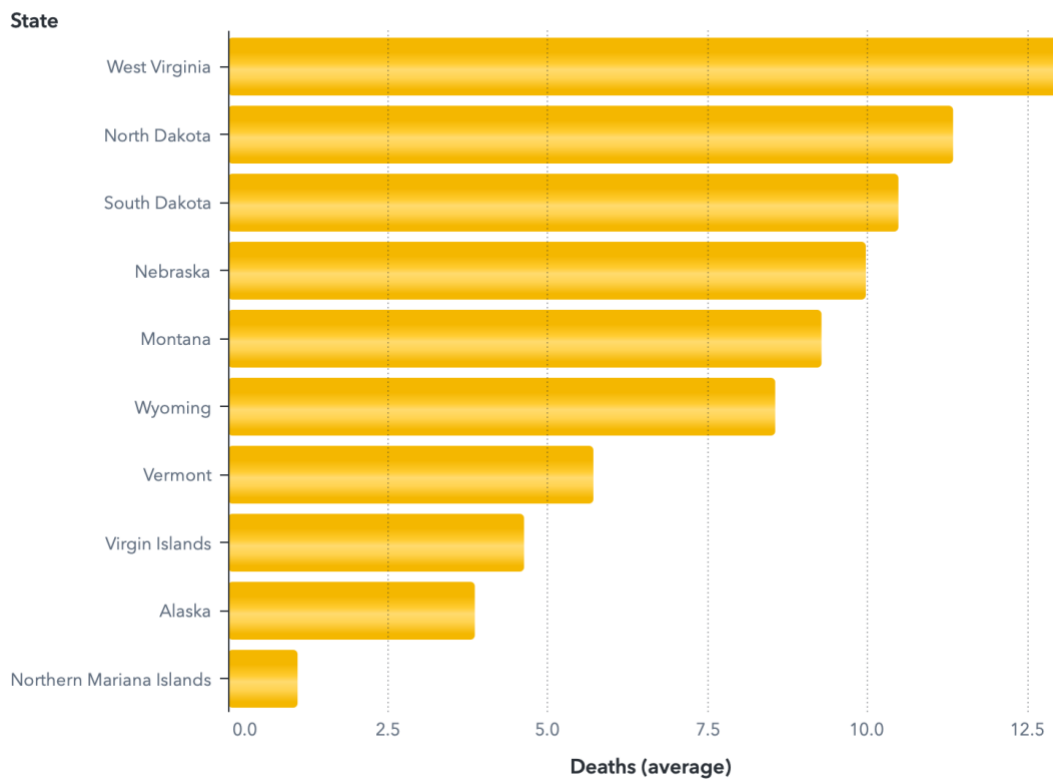


Figure 5.4

Finally, the past COVID-19 trends suggested that there might be a significant increase of COVID-19 cases across the US within the next 12 months (*see Figure 6.1*). Moreover, this country was responsible for the greatest number of COVID-19 cases and fatalities worldwide (Worldometers 2024). Hence, it seems that the US will experience a large growth of COVID-19 cases in the future, considering the previous 12 months. Nevertheless, I found that there will be a slight increase in COVID-19 cases when using SAS Viya to forecast the COVID-19 dataset for the upcoming 12 months (*shown on Figure 6.2*). The COVID-19 cases across the US will be increased by 13.39% (Sas Viya for Learning 2024). Thereby, the US will experience an increase of COVID-19 cases, but it will not skyrocket. In spite the relatively low percentage of the predicted “increase” of COVID-19 cases, the predicted number of an increase is still quite high, 86 million (Sas Viya for Learning 2024). Therefore, I suggest the US public healthcare to further mitigate the transmission of COVID-19 to reduce the possibility of further transmission of the virus.

Cases by Date

Cases (millions)

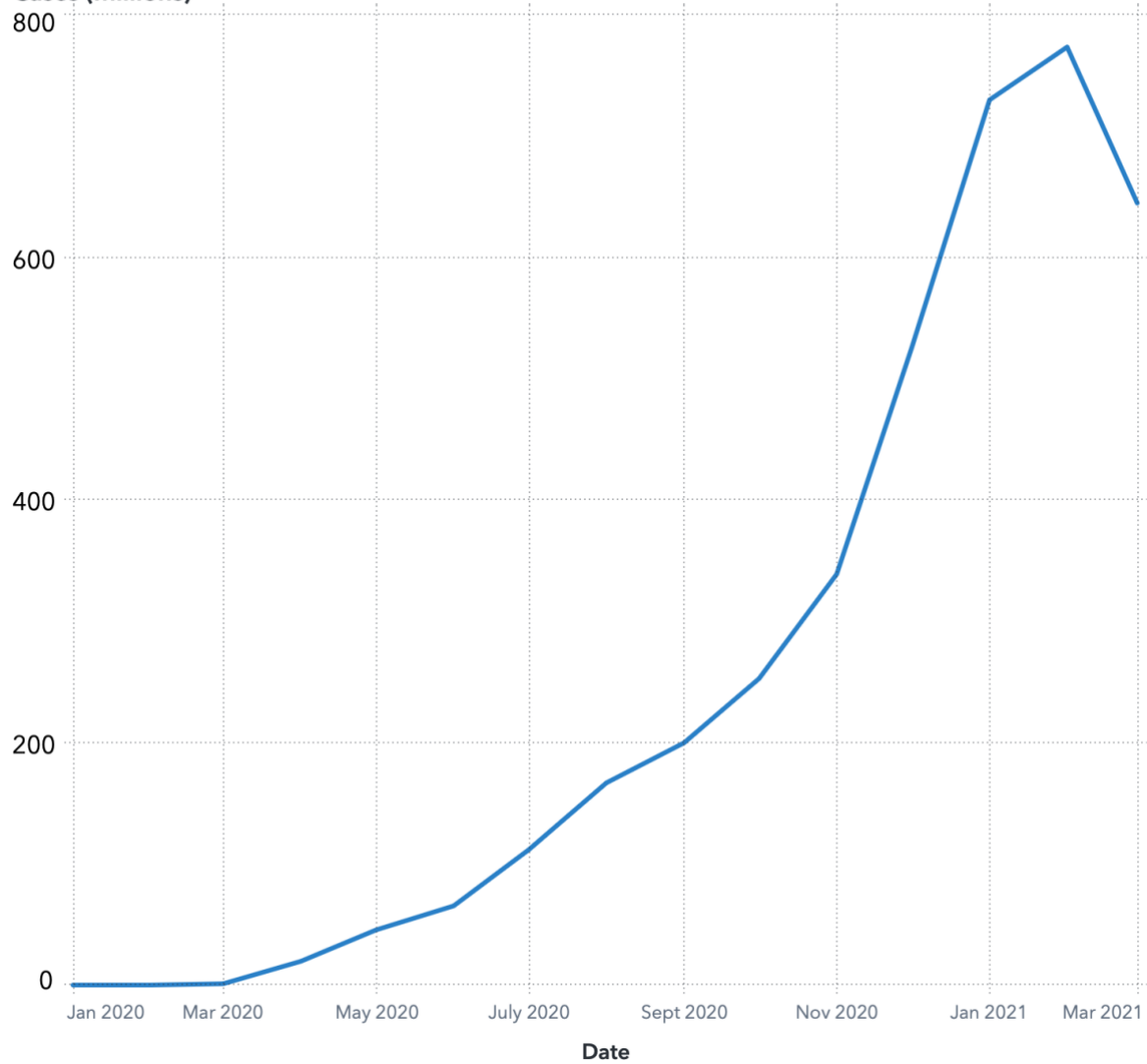


Figure 6.1

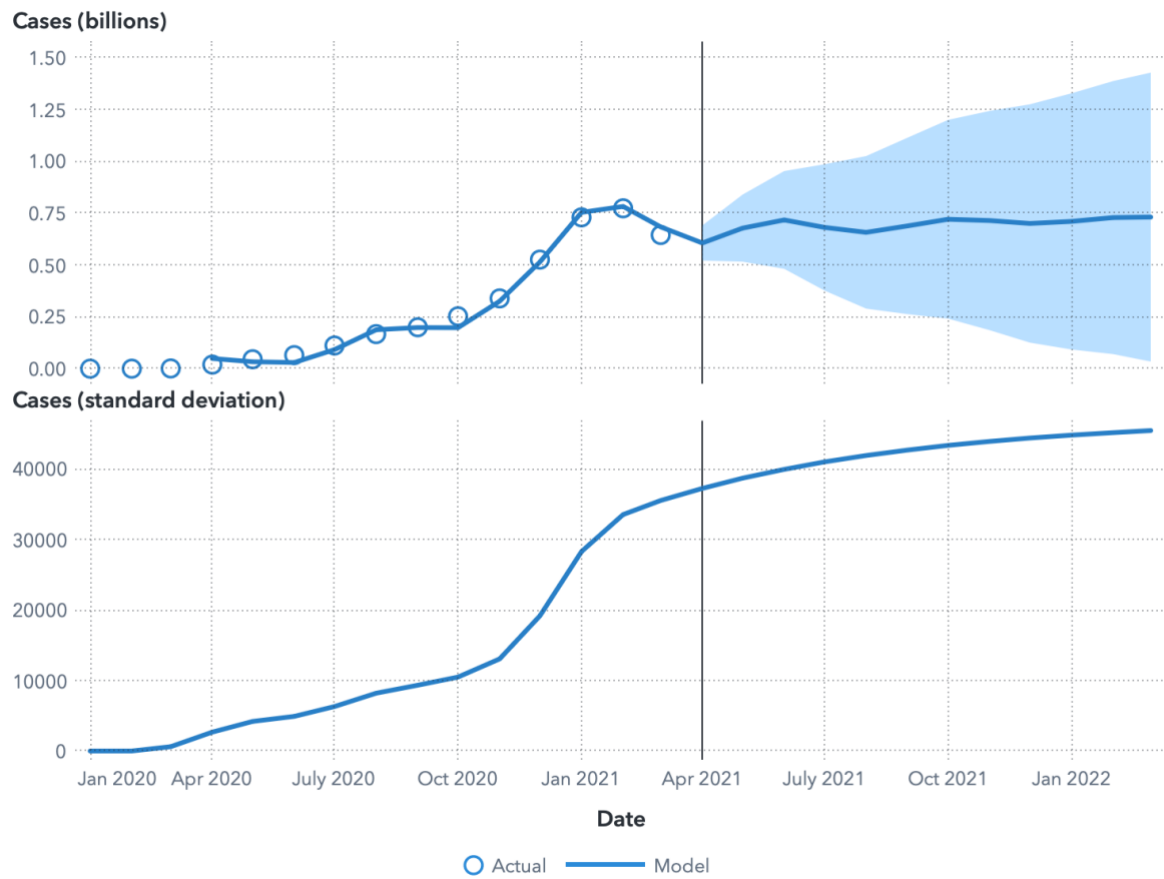


Figure 6.2

Conclusion

Summary

To lessen the negative impacts, the US government and health organisations decided to monitor it. The NYT has published the COVID-19 dataset containing the cases and deaths across the US from January 2020 to March 2021. The insights I received from the dataset are written on this analysis. The primary objective of this analysis is to identify the trends and patterns of COVID-19 and generate COVID-19 information. This will assist the public healthcare generate insightful decisions in mitigating the pandemic's effect through the effective written communication regarding the COVID-19 trends.

Key Insights

The following key insights are used to develop this COVID-19 analysis to meet the overall aims purposed to effectively communicate the COVID-19 dynamics to the public health:

1. The correlation between the cases and causalities.
2. The comparison between the total and average cases and deaths by state.
3. The table list that categorises counties with highest cases by state.
4. The States with highest and lowest COVID-19 cases.
5. The forecast to predict the trends of COVID-19 cases within 12 months.

Recommendations

To alleviate the pandemic, I suggest the following recommendations for the US public health care:

1. Vaccination mandatory in the most affected states and counties.

To alleviate the further transmission, particularly in the ten states and counties that experience the highest COVID-19 cases and deaths (in both average and total sum), I highly recommend the vaccination policy. This is since the COVID-19 vaccination initiatives promote the high coverage and improve immune system to prevent the severity of coronavirus in our body (Moghadas et al. 2021, pp. 6-7) (*refer to Figure 7 in Appendix*). For the COVID-19 vaccination in the US, it is recommended to prioritise ten most affected states and counties, followed by the states and counties with higher cases and causalities among the regions with rarer cases and deaths.

2. Travel and movement restriction in the given time.

Until there is a significant decline in the COVID-19 cases and mortality rate, I suggest the lockdown implementation to restrict travel and mobility. In addition to lockdown, the US citizens who reside in the states and counties with highest cases, alongside

with higher mortality rate due to the COVID-19 should implement home quarantine to further prevent the transmission of the virus (Talic et al. 2021, pp. 9-10). Furthermore, I suggest the implementation of this effort to be applied in the states and counties with highest rate of COVID-19 cases and mortality rate (in both total and average measurement). This is to further decrease the COVID-19 cases and fatalities in the most affected regions in the US.

Reference List

BBC 2020, *Covid-19: Milestones of the global pandemic*, BBC, viewed 22 September 2024, <<https://www.bbc.com/news/world-54337098>>.

BBC 2022, *Covid map: Coronavirus cases, deaths, vaccinations by country*, BBC, viewed 22 September 2024, <<https://www.bbc.com/news/world-51235105>>.

Hale, E 2023, *Three years later, what we know: How did COVID start?*, Al Jazeera, viewed 22 September 2024, <<https://www.aljazeera.com/features/2023/3/11/three-years-later-what-we-know-how-did-covid-19-start>>.

https://www.sas.com/en_au/software/viya-for-learners.html

Moghadas, SM, Vilches, TN, Zhang, K, Wells, CR, Shoukat, A, Singer, BH, Meyers, LA, Neuzil, KM, Langley, JM, Fitzpatrick, MC & Galvani, AP 2021, 'The impact of vaccination on COVID-19 outbreaks in the United States', *MedRxiv: The Preprint Server for Health Sciences*, pp. 6-7; 16.

SAS Viya for Learning. (2024). Retrieved from

Talic, S, Shah, S, Wild, H, Gasevic, D, Maharaj, A, Ademi, Z, Li, X, Xu, W, Mesa-Equiagaray, I, Rostron, J, Theodoratou, E, Zhang, X, Motee, A, Liew, D & Ilic, D 2021, 'Effectiveness of public health measures in reducing the incidence of covid-19, SARS-CoV-2 transmission, and covid-19 mortality: systematic review and meta-analysis', *The BMJ*, pp. 9-10.

Wisevoter 2023, *States with the most counties*, Wisevoter, viewed 25 September 2024, <<https://wisevoter.com/state-rankings/states-with-the-most-counties/>>.

World Health Organization 2020, *Tracking COVID-19: contact tracing in the digital age*, World Health Organization, viewed 22 September 2024, <<https://www.who.int/news-room/feature-stories/detail/tracking-covid-19-contact-tracing-in-the-digital-age>>.

World Health Organization 2021, *Contact tracing in the context of COVID-19*, World Health Organization, viewed 22 September 2024, <https://iris.who.int/bitstream/handle/10665/339128/WHO-2019-nCoV-Contact_Tracing-2021.1-eng.pdf>.

World Population Review 2024, *States with the most counties 2024*, World Population Review, viewed 24 September 2024,

<<https://worldpopulationreview.com/state-rankings/county-count-by-state>>.

Worldometers 2024, *Countries where COVID-19 has spread*, Worldometers, viewed 26 September 2024,

<<https://www.worldometers.info/coronavirus/countries-where-coronavirus-has-spread/>>.

Appendix

Column Name	Data Type	Description
Date	Date	The date of the record in YYYY-MM-DD format.
County	String	The name of the county.
State	String	The name of the state.
FIPS	Float	The Federal Information Processing Standards code for the county.
Cases	Integer	The total number of confirmed COVID-19 cases in the county on the specified date.
Deaths	Integer	The total number of confirmed COVID-19 related deaths in the county on the specified date.
statename	String	The name of the state (repeated for redundancy).
stat_num	Float	A numerical code representing the state.
STATECODE	String	The abbreviated code for the state.
COUNTRY	Float	The numerical code for the country (926.0 represents the USA).
id	Integer	A unique identifier for the record.
state_code	String	The abbreviated code for the state (repeated for redundancy).
state_id	String	The unique identifier for the state (e.g., US-17 for Illinois).
county_id	String	The unique identifier for the county within the state.
COUNTYCODE	String	The county code.
IDNAME	String	The name of the county (repeated for redundancy).
county_num	Float	A numerical code representing the county.
state_id	String	The unique identifier for the state (repeated for redundancy).
state_num	Float	A numerical code representing the state (repeated for redundancy).
county_Latitude	Float	The latitude coordinate of the county.
county_Longitude	Float	The longitude coordinate of the county.
County_Longitude_X	Float	The transformed longitude coordinate for mapping purposes.
County_Latitude_Y	Float	The transformed latitude coordinate for mapping purposes.
country_id	String	The unique identifier for the country (US for the United States).
country_state_id	String	The unique identifier combining the country and state (e.g., US for the United States).

Data Dictionary (UOW 2024)

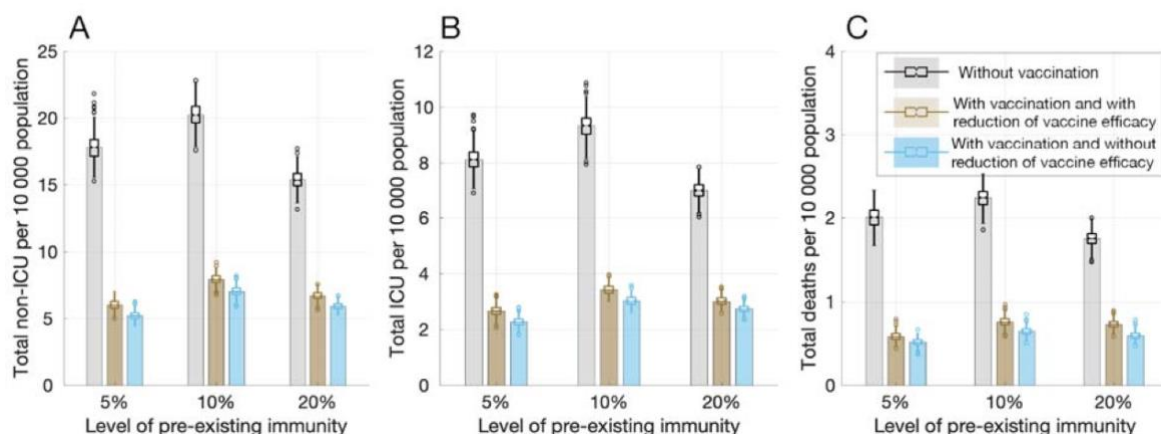


Figure 7 (Moghadas et al. 2021, p. 16)